

4

Data Visualization

Business Analytics, 2e

By Sanjiv Jaggia, Alison Kelly, Kevin Lertwachara, and Leida Chen

Chapter 4 Learning Objectives (LOs)

LO 4.1 Visualize a single variable.

LO 4.2 Visualize the relationship between two variables.

LO 4.3 Visualize the relationship between two or more variables.

Introductory Case: Construction Clothing

- ReliableWorkWear.com is an online company that offers a large selection of construction clothing, work boots, gloves, and more that keep workers safe and comfortable on the job.
- Brendan Moore is the marketing analyst for ReliableWorkWear.com. He has compiled data on 200 recent transactions

Customer	Purchase	Satisfaction	Repeat	Sex	Income
1	241	1	No	Male	53
2	843	3	Yes	Male	94
⋮	⋮	⋮	⋮	⋮	⋮
200	926	4	Yes	Male	102

- Brendan will use the sample information to:
 1. Convey the information from the variables in tabular form.
 2. Convey the information from the variables in graphical form.
 3. Discuss finding and provide strategies that may help increase sales.

Methods to Visualize a Single Variable

- **Tabular** and **graphical** tools help us to organize and present data.
- We can summarize both a *categorical variable* as well as a *numerical variable*.
- A **categorical variable** consists of observations that represent labels or names.
 - it is often ***useful to summarize the variable with a frequency distribution and/or a bar chart***.

Methods to Visualize a Single Variable (Category)

- Example: Consider the Repeat and Sex variables from the introductory case.
- With Excel
 - Use COUNTIF
 - Insert > Insert Column or Bar Chart > 2-D Column
 - Add Chart Elements > Data Labels > Outside End

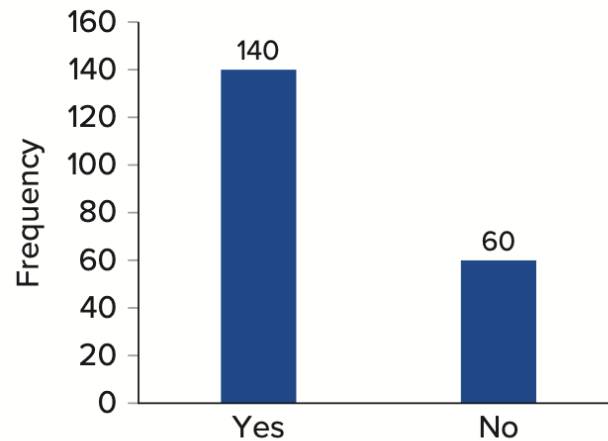
Methods to Visualize a Single Variable

- Example continued

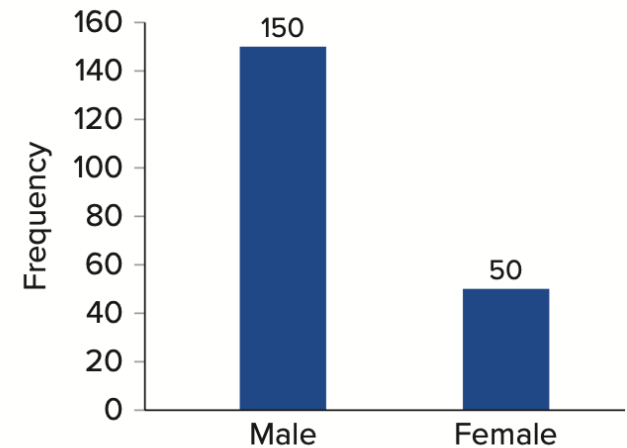
(a) The Repeat Variable		
Repeat	Frequency	Relative Frequency
Yes	140	0.70
No	60	0.30

(b) The Sex Variable		
Sex	Frequency	Relative Frequency
Male	150	0.75
Female	50	0.25

(a) The Repeat variable



(b) The Sex variable



Methods to Visualize a Single Variable (Numeric)

- For a numerical variable, a frequency distribution groups the observations into intervals and records the number of observations that falls into each interval.
 - The relative frequency for each interval equals the proportion of observations in each interval.
- The data are more manageable using a frequency distribution, but some detail is lost.
 - Histogram

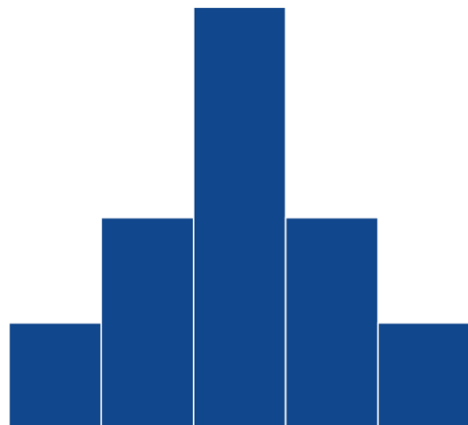
Visualize a Single Variable: Histogram

- A histogram is a series of rectangles (bar) where the width and height of each rectangle represent the interval width and frequency (or relative frequency) of the respective interval (binning).
 - The height of each bar represents either the frequency or the relative frequency for each interval.

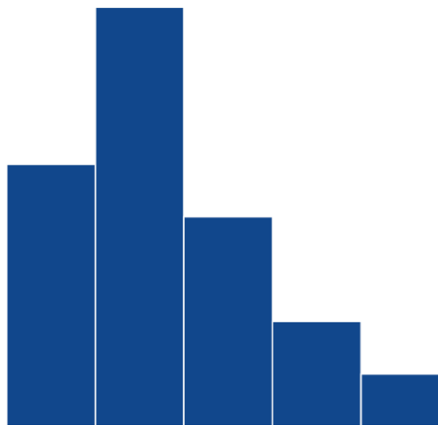
Visualize a Single Variable: Histogram

- A histogram provides information about the shape of the distribution.
 - **Symmetric:** mirror image of itself on both sides of its center
 - **Skewed:** positive (elongated right tail) or negative (elongated left left)

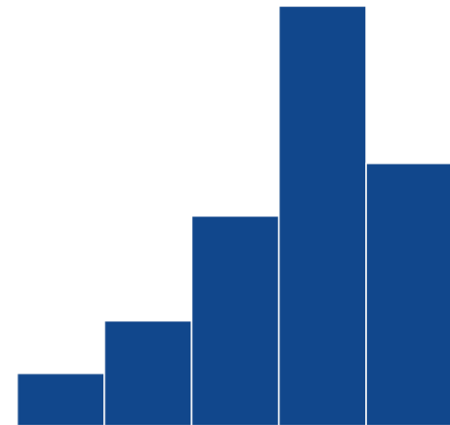
(a) Symmetric distribution



(b) Positively skewed distribution



(c) Negatively skewed distribution

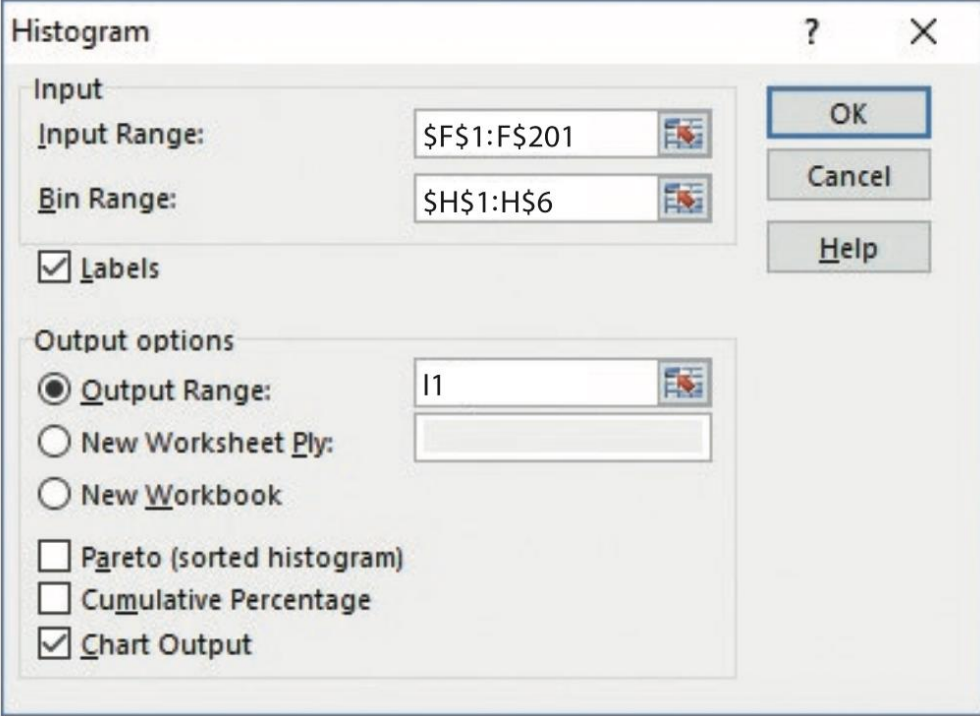


Visualize a Single Variable: Histogram

- We must make certain decisions about the number of intervals, as well as the width of each interval.
 - The intervals are ***mutually exclusive***.
 - The total number of intervals usually ranges from 5 to 20.
 - The ***intervals are exhaustive*** (contains all the values of the variables.)

Methods to Visualize a Single Variable

- Example: The Income variable from the introductory case.
- With Excel: Data > Data Analysis > Histogram



The image shows the 'Histogram' dialog box in Microsoft Excel. The dialog box has a title bar with a question mark and a close button. It contains several input fields and checkboxes. The 'Input' section has 'Input Range' set to '\$F\$1:F\$201' and 'Bin Range' set to '\$H\$1:H\$6'. The 'Labels' checkbox is checked. The 'Output options' section has 'Output Range' set to 'I1', with 'New Worksheet Ply' and 'New Workbook' as alternative options. The 'Pareto (sorted histogram)' and 'Cumulative Percentage' checkboxes are unchecked, while the 'Chart Output' checkbox is checked. On the right side of the dialog box, there are three buttons: 'OK', 'Cancel', and 'Help'.

Histogram

Input

Input Range: \$F\$1:F\$201

Bin Range: \$H\$1:H\$6

☒ Labels

Output options

☒ Output Range: I1

☐ New Worksheet Ply:

☐ New Workbook

☐ Pareto (sorted histogram)

☐ Cumulative Percentage

☒ Chart Output

OK

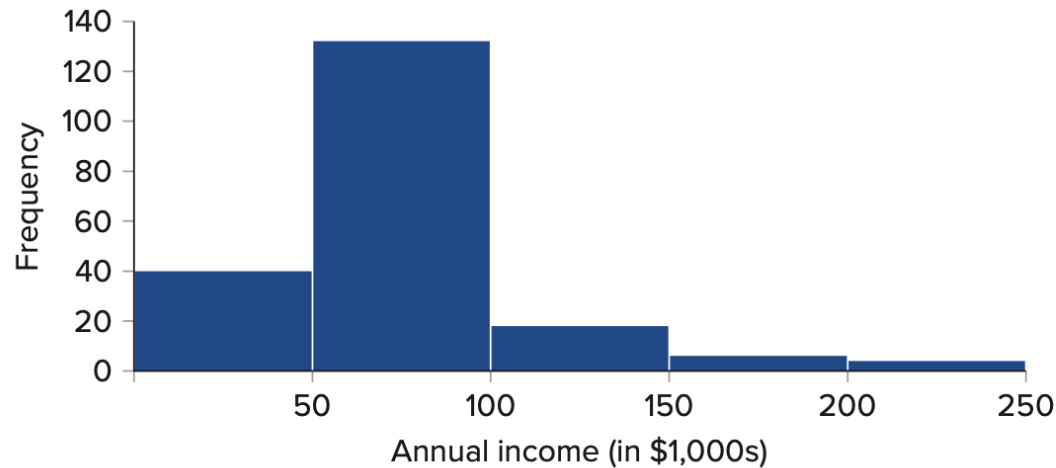
Cancel

Help

Methods to Visualize a Single Variable

- Example continued

Interval (in \$1,000s)	Frequency	Relative Frequency
$0 < x \leq 50$	40	0.20
$50 < x \leq 100$	132	0.66
$100 < x \leq 150$	18	0.09
$150 < x \leq 200$	6	0.03
$200 < x \leq 250$	4	0.02



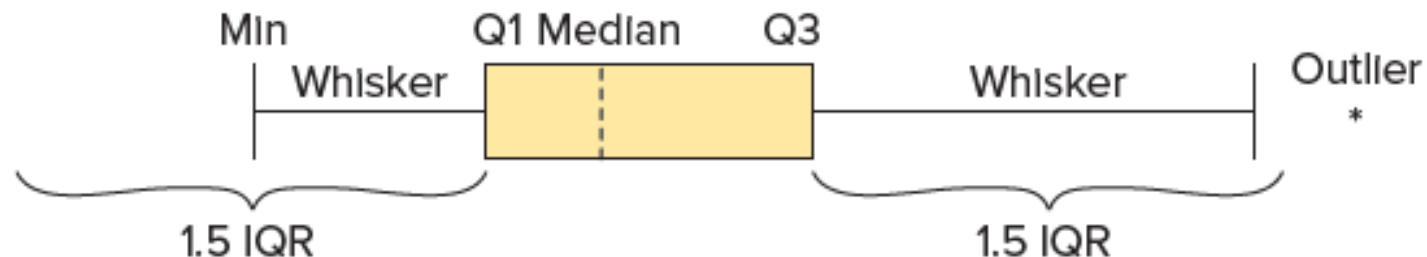
Revisit: Investment Decision

- The investor shows Dorothy the annual return data for Fidelity's Growth Index Fund (Growth) and Fidelity's Value Index fund.

Year	Growth	Value
1984	−5.50	−8.59
1985	39.91	22.10
⋮	⋮	⋮
2019	38.42	31.62

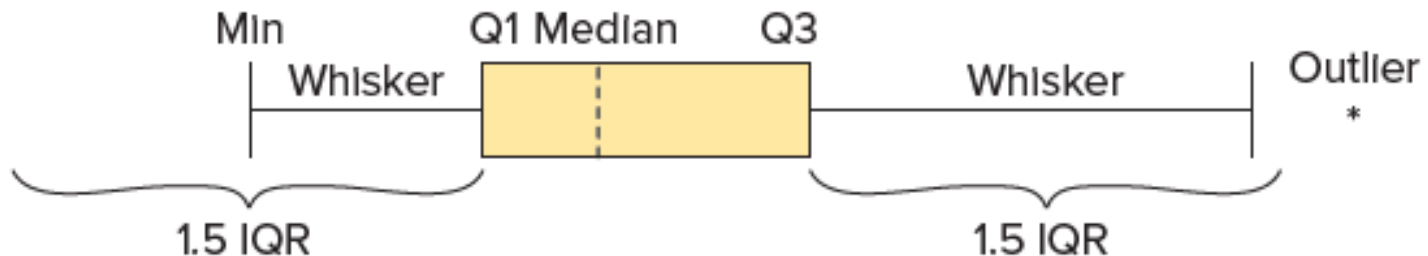
Boxplot: Visualize a single variable

- A boxplot, also referred to as a **box-and-whisker plot**, is a way to graphically **display a five-number summary** for a single variable.
 - the minimum, the quartiles [Q1, Q2 (median), and Q3], and the maximum.
 - Draw a box encompassing the first and third quartiles.
 - Draw a dashed vertical line in the box at the median.
 - Calculate the **IQR**. Draw a whisker that extends from Q1 to the minimum value that is not further from $1.5 \times \text{IQR}$ from Q1. Similarly, draw a line that extends from Q3 to the maximum value that is not farther than $1.5 \times \text{IQR}$ from Q3.
 - Use an asterisk (or another symbol) to indicate observations that are farther than $1.5 \times \text{IQR}$ from the box. These observations are considered **outliers**.



Analysis using Boxplot

- A boxplot is also used to informally gauge the **shape** of the distribution.
- **Symmetry** is implied if the median is in the center of the box and the left/right whiskers are equidistant from their respective quartiles.
- If the median is left of center and the right whisker is longer than the left whisker, then the distribution is **positively skewed**.
- Similarly, if the median is right of center and the left whisker is longer than the right whisker, then the distribution is **negatively skewed**.
- If **outliers** exist, we need to include them when comparing the lengths of the left and right whiskers.



Methods to Visualize the Relationship between Two Variables

- Use a **contingency table** to summarize and examine the relationship between two categorical variables.
 - Frequencies for two categorical variables
 - Each cell represents a mutually exclusive combination of the pair of values
- The contingency table allows us to present and interpret the raw data in a much more manageable format.

Methods to Visualize the Relationship between Two Variables

The information in a contingency table can be shown graphically using **a stacked column chart**.

- It is designed to visualize more than one categorical variable.
- It allows for the comparison composite within each category.

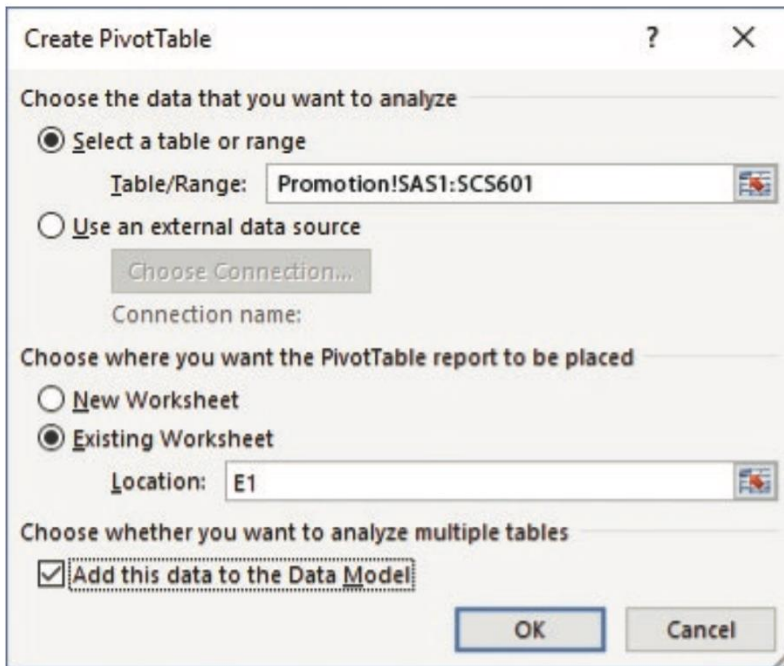
Methods to Visualize the Relationship between Two Variables

- Example: An online retailer recently sent e-mails to customers that included a promotional discount.
- The retailer wonders whether there is any relationship between a customer's location in the U.S. and whether the customer made a purchase with the discount.

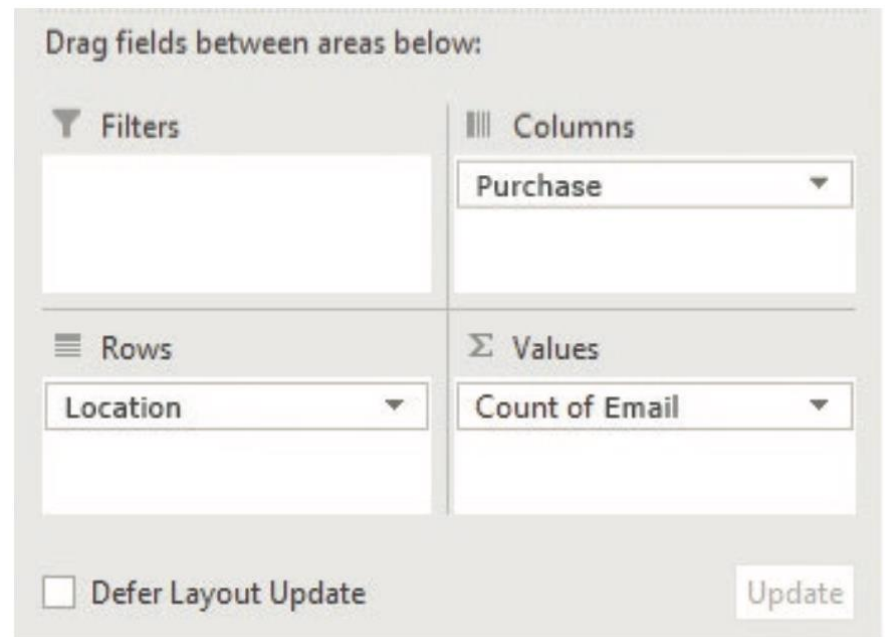
Email	Location	Purchase
1	West	yes
2	Northeast	yes
⋮	⋮	⋮
600	South	no

Methods to Visualize the Relationship between Two Variables

- Example continued
- With Excel: Insert > Pivot Table



The 'Create PivotTable' dialog box is shown. It has a title bar with a question mark and a close button. The main area is divided into three sections. The first section, 'Choose the data that you want to analyze', has two radio buttons: 'Select a table or range' (selected) and 'Use an external data source'. Below the first radio button is a text box labeled 'Table/Range:' containing 'Promotion!\$A\$1:\$C\$601'. Below the second radio button is a 'Choose Connection...' button and a 'Connection name:' text box. The second section, 'Choose where you want the PivotTable report to be placed', has two radio buttons: 'New Worksheet' and 'Existing Worksheet' (selected). Below the second radio button is a 'Location:' text box containing 'E1'. The third section, 'Choose whether you want to analyze multiple tables', has a checked checkbox labeled 'Add this data to the Data Model'. At the bottom are 'OK' and 'Cancel' buttons.



The PivotTable field list task pane is shown. It has a title bar that says 'Drag fields between areas below:'. The main area is divided into four sections: 'Filters' (empty), 'Columns' (containing a dropdown menu with 'Purchase' selected), 'Rows' (containing a dropdown menu with 'Location' selected), and 'Values' (containing a dropdown menu with 'Count of Email' selected). At the bottom is a checkbox labeled 'Defer Layout Update' and an 'Update' button.

Methods to Visualize the Relationship between Two Variables

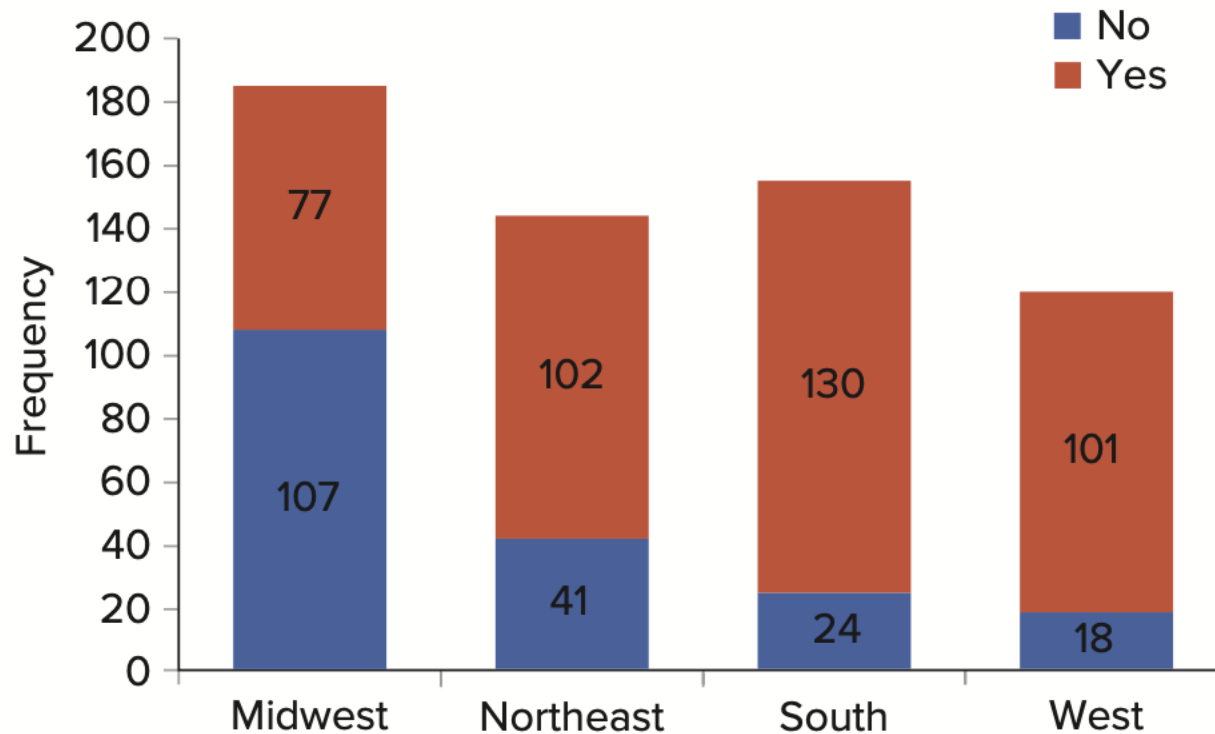
- Example continued with Excel: **Contingency table**

Location	Purchase		Total
	No	Yes	
Midwest	107	77	184
Northeast	41	102	143
South	24	130	154
West	18	101	119
Total	190	410	600

Location	Purchase		Total
	No	Yes	
Midwest	17.83%	12.83%	30.67%
Northeast	6.83%	17.00%	23.83%
South	4.00%	21.67%	25.67%
West	3.00%	16.83%	19.83%
Total	31.67%	68.33%	100.00%

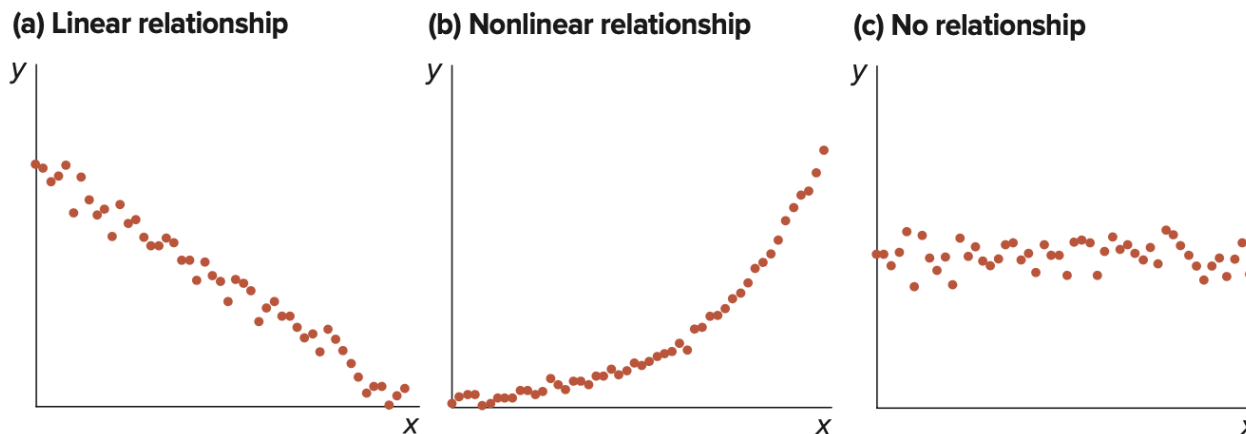
Methods to Visualize the Relationship between Two Variables

- Example continued with Excel
- Insert > Insert Column or Bar Chart > Stacked Column



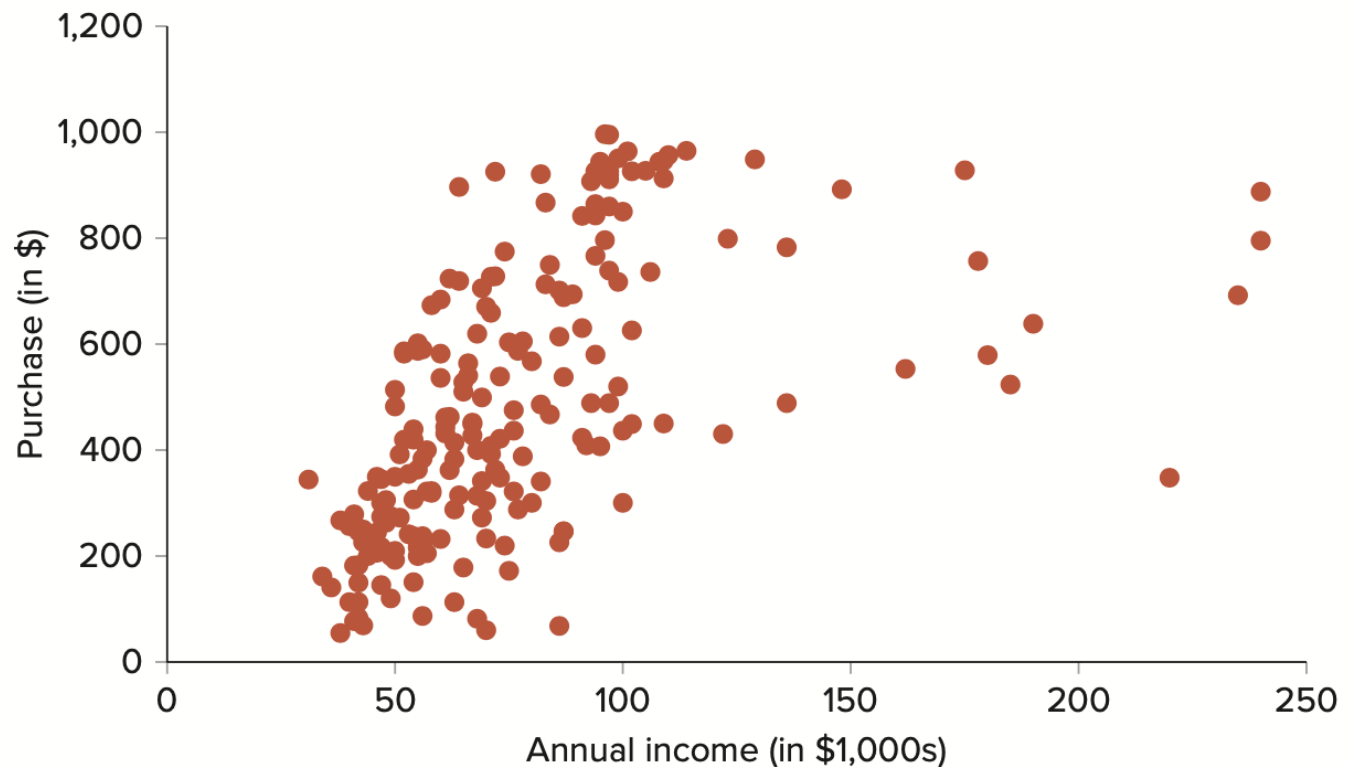
Methods to Visualize the Relationship between Two Variables

- A **scatterplot** is used to determine whether or not two numerical variables are related in some systematic way.
 - Each point represents a paired observation for the two variables
 - Refer to one variable as x (x-axis) and the other as y (y-axis)
- Once plotted, the graph may reveal one of the below.
 - A linear relationship
 - A nonlinear relationship
 - No relationship



Methods to Visualize the Relationship between Two Variables

- Example: Consider the customer's purchase amount and annual income.
- With Excel: Insert > Insert Scatter or Bubble Chart > Scatter



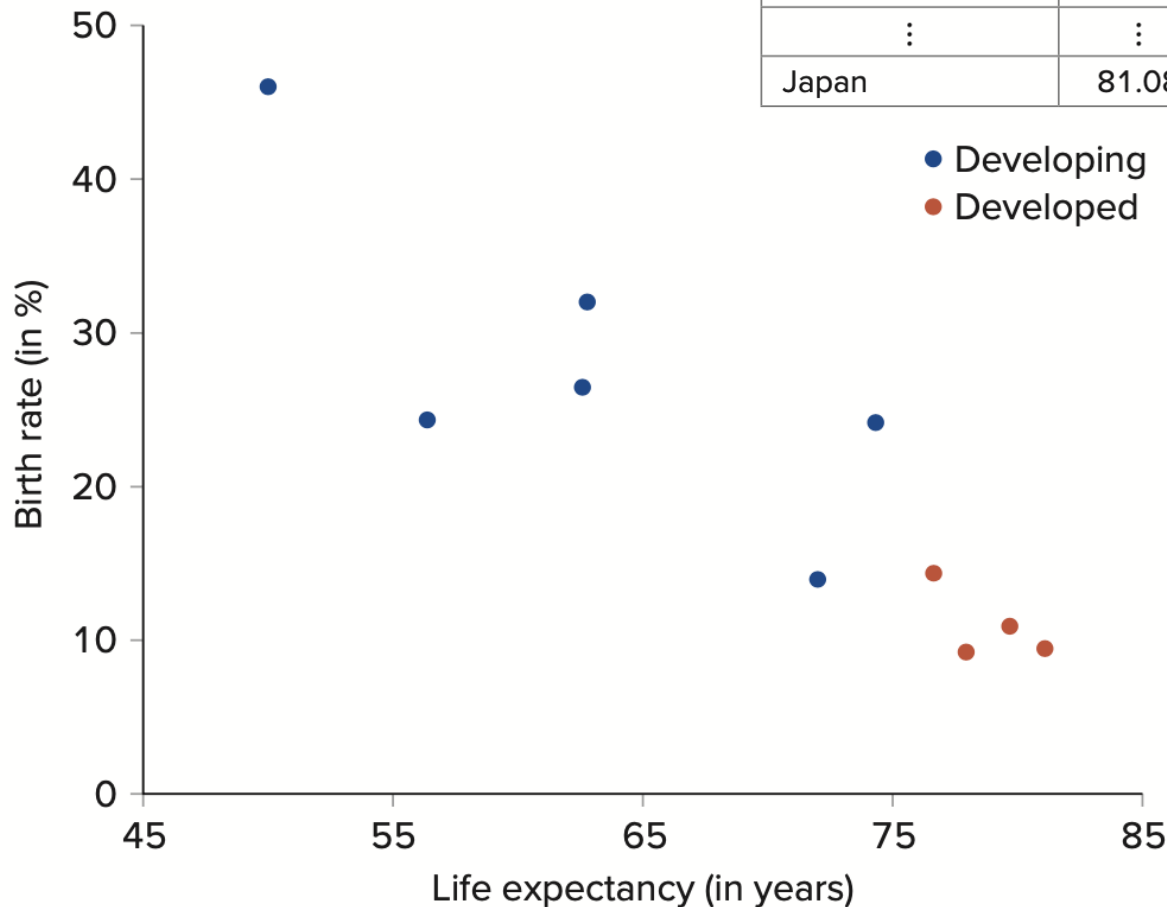
Scatter plot

- A **scatterplot** with a categorical variable modifies a basic scatterplot by incorporating a categorical variable.
- This allows us to see if the relationship between two numeric variables differs across the levels of a categorical variable.
- It is **common to encode the categorical variable through point color**.
 - Giving each point a distinct hue makes it easy to show its membership to a respective category.

Scatter plot: Using color hue

- Example: Life expectancy vs. birth rate by country

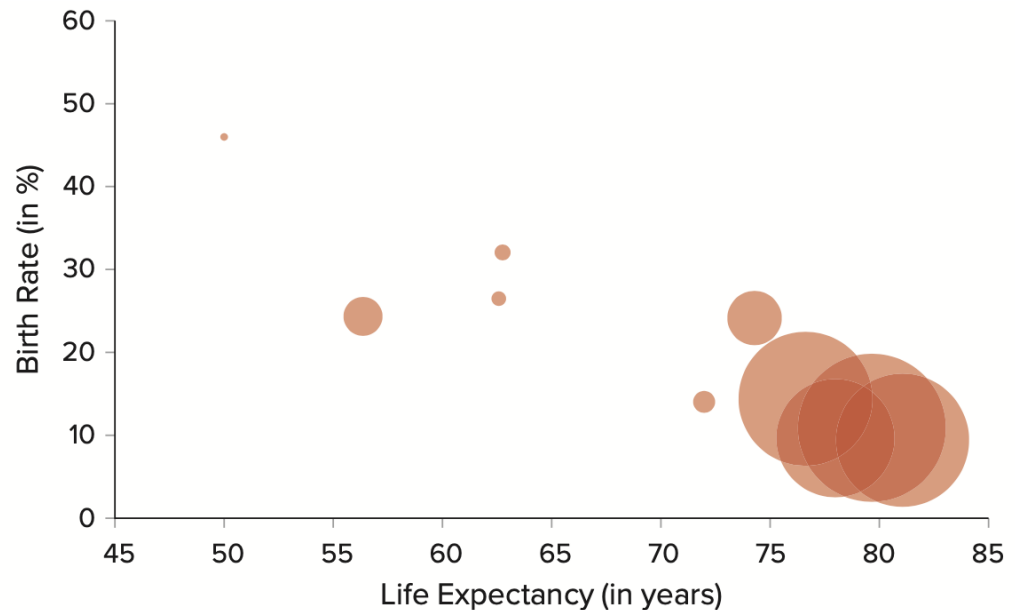
Country Name	Life_Exp	Birth_Rate	GNI	Development
Congo, Dem. Rep.	50.00	45.96	130	Developing
India	62.59	26.46	440	Developing
⋮	⋮	⋮	⋮	⋮
Japan	81.08	9.40	36230	Developed



With Excel: Insert > Insert Scatter or Bubble Chart > Scatter; add Edit Series

Scatter: Bubble plot

- A **bubble plot** shows the relationship between three numerical variables.
- The third variable is represented by the size of the bubble (points).
- Example: life expectancy vs. birth rate by country development
- With Excel: Insert > Insert Scatter or Bubble Chart > Bubble



Line Chart

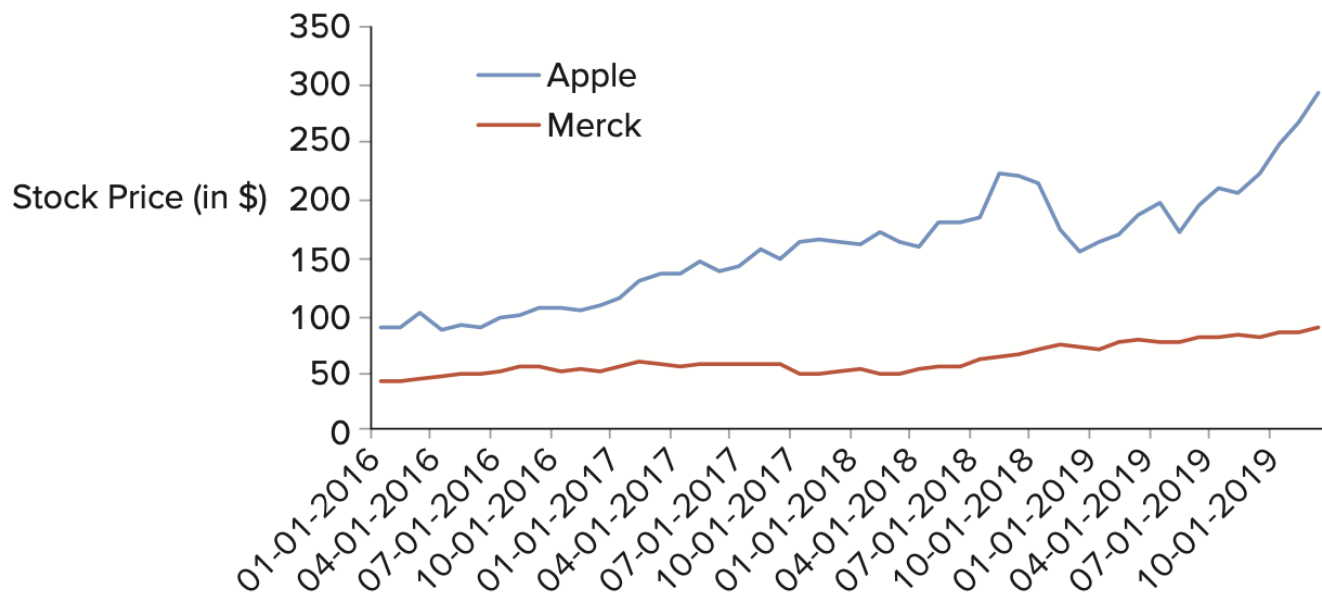
- A **line chart** displays a numerical variable as a series of data points connected by a line.
 - Connects the **consecutive observations of a numerical variable** with a line.
 - A line chart is especially **useful for tracking changes** or trends over time.
- It is also easy for us to identify any major changes that happened in the past on a line chart.
- When multiple lines are plotted in the same chart, we can compare these observations on one or more dimensions.

4.3: Other Data Visualization Methods

- Example: Monthly stock prices for Apple and Merck.

Date	Apple	Merck
1/1/2016	90.96	44.98
2/1/2016	90.35	44.57
⋮	⋮	⋮
12/1/2019	293.65	90.33

- With Excel: Insert > Insert Line or Area Chart > Line



Conclusion

- A set of common visualizations for categorical and numerical variables are introduced.
 - For single variable, using bar and histogram, Box plot and line chart
 - For multiple variables, many possible techniques, e.g. stacked bars, scatter plot
- Special form of visualization to express relationship
 - Scatter with colors, bubble
 - Colored line chart